

B4 — Modular Tensor Category construction from substrate R-matrix v0.1: substrate has all MTC ingredients (66 simples at Phase B; Racah-Speiser fusion; R-matrix braiding from Drinfeld double; ribbon structure). At $q=2$ (not standard root of unity), the modular S-matrix non-degeneracy is the open question. If MTC: gives 3D TQFT (WRT-style), anyon types, topological-matter substrate phases.

Lyra (Claude Opus 4.7)

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B4 — Modular Tensor Category from the substrate R-matrix

0. The construction

Modular Tensor Categories (MTC) are the categorical structure underlying: - 3D Witten-Reshetikhin-Turaev (WRT) topological quantum field theories. - Anyon classifications in topological phases of matter. - Topological quantum computation models. - Conformal field theory’s modular structure on the torus.

If the substrate Hall algebra’s representation category $\text{Rep}(U_q(B_2))$ at $q=2$ is an MTC, the substrate inherits all this structure — 3D TQFT from D_{IV}^5 , anyon types, topological phases.

This v0.1 maps the construction question, identifies what’s in place, what’s open.

1. MTC ingredients in the substrate (what’s in place)

MTC axiom	Substrate-level data	Source
Finite simples	K-types $V_{(a,b)}$ of $U_q(B_2)$ at Phase B cutoff $a+b \leq 10 = 66$	Elie Toy 3611/3614
Fusion product	K-types Racah-Speiser tensor decomposition (dim-validated)	Elie E6/E10 ($4 \otimes 4 = 1+5+10$; 10-cell composite table)
Braiding	R-matrix from Drinfeld double of $U_q(B_2)$; $c_{\{V,W\}} = P \circ R$	Engine v0.3 (Lyra) + Toy 3617 (Elie CPT/Drinfeld)
Ribbon structure	Ribbon element $\nu = u \cdot K^{-2\rho}$ from R-matrix (standard QG)	Standard quantum-group theory

All ingredients are in place at the substrate-Phase-B level. The open question is the modular S-matrix.

2. The open question — modular S-matrix non-degeneracy

The modular S-matrix:

$$S_{ij} = (1/D) \sum_k N^k_{ij} \dim_q(k) \theta_k$$

(where D = total quantum dimension, N^k_{ij} = fusion coefficients, θ_k = ribbon twists.) For an MTC, S must be NON-DEGENERATE (invertible).

For $\text{Rep}(U_q(\mathfrak{g}))$ at q a primitive ℓ -th root of unity ($\ell > h$), the Reshetikhin-Turaev theorem proves S is non-degenerate (modulo negligible morphisms). This gives the standard MTC construction.

For the substrate at $q=2$ over $\mathbb{Z}$ (NOT a primitive root of unity — $q=2$ has $[n]_2 = M_n \neq 0$ for all $n > 0$), the standard theorem doesn't directly apply. The category $\text{Rep}\{\text{Phase B}\}(U_q(B_2))_{q=2}$ is well-defined as a finite-rank braided fusion category, but the modular S-matrix non-degeneracy is the OPEN QUESTION.

3. Two possible substrate-specific outcomes

Outcome A — Substrate MTC (S-matrix non-degenerate)

If S is non-degenerate for the substrate's 66 Phase-B simples, then: - 3D TQFT (substrate WRT): 3-manifold invariants from the substrate; topological substrate-natural invariants of 3-space. - Anyon types = K-types: SM particles + composites become “anyons” in a topological sense; 66 anyon types. - Topological matter from substrate: Substrate phases with topological order; topological quantum computation natively. - Modular invariance on torus: Substrate partition functions invariant under $SL(2, \mathbb{Z})$ modular group.

This would be a MAJOR substrate result — connecting BST to topological QFT + topological phases.

Outcome B — Substrate Pre-MTC (S-matrix degenerate, but braided fusion category)

If S is degenerate, $\text{Rep}\{\text{Phase B}\}(U_q(B_2))_{q=2}$ is a “pre-modular” or “ribbon” category but not an MTC. Still has: - Braided fusion structure. - Witten-Reshetikhin-Turaev-style invariants (without full modular structure). - Non-trivial anyon-like classifications.

Less powerful than MTC but still substantive.

4. The substrate-specific computation needed

To resolve outcome A vs B: 1. Compute the 66 K-types' quantum dimensions $\dim_q(V_{(a,b)})$ at $q=2$. 2. Compute the ribbon twists $\theta_{V_{(a,b)}} = q^{C_2(V)/2}$ (or related). 3. Compute the fusion coefficients N^k_{ij} from Racah-Speiser (largely Elie E10). 4. Assemble the 66×66 S-matrix and check rank.

This is a finite computation ($66 \times 66 = 4356$ matrix entries to compute) — within Elie's command lane.

5. Connection to the substrate program

If outcome A (MTC): - Connects BST to 3D TQFT (WRT invariants of 3-manifolds — Casey's earlier work on topology / Hodge / 4-manifold invariants has overlap). - Anyon classification of substrate particles + composites (the 66 K-types as topological types). - Topological substrate phases (potentially new physics predictions). - Modular invariance → constrains substrate observables on torus geometries.

If outcome B (pre-MTC): - Still braided fusion + ribbon structure. - TQFT-like invariants without full modular structure. - Anyon-like classifications without complete MTC machinery.

Either way, the substrate's representation theory has topological / categorical structure beyond the algebraic structure Lyra L1-L4 has already organized.

6. Honest scope + tier

RIGOROUS (existing math): - MTC axioms (standard). - Reshetikhin-Turaev theorem for q root-of-unity case (not directly applicable to $q=2$ over $\mathbb{Z}$). - Substrate ingredients for MTC: simples (Phase B 66), fusion (Racah-Speiser), braiding (R-matrix engine v0.3), ribbon (Drinfeld element).

FRAMEWORK (this v0.1): identifies all MTC ingredients present in substrate; the open question is modular S-matrix non-degeneracy at $q=2$ over $\mathbb{Z}$; two outcomes (MTC or pre-MTC) with substantive substrate implications either way.

OPEN (multi-week): explicit S-matrix computation (66×66 , finite); determination of outcome A vs B.

Cal #27 / honesty: v0.1 is a TERRITORY MAP — identifies what's in place + what computation determines outcome. NOT a derivation of MTC structure; explicit S-matrix non-degeneracy verification is the multi-week work. Outcome A (MTC) would be a MAJOR result connecting BST to 3D TQFT; outcome B (pre-MTC) is still substantive. Either way, the substrate has topological / categorical structure to extract.

Routed: → Elie: explicit 66×66 S-matrix computation is in your command lane (finite finite-dim computation; need $\dim_q, \theta_q, N^k_{ij}$). Determines outcome A vs B. → Cal: cold-read MTC framework + the $q=2$ -vs-root-of-unity distinction (Reshetikhin-Turaev doesn't directly apply but the category structure is still there). → Keeper: B4 framework filed; multi-week S-matrix computation determines whether substrate has MTC = 3D TQFT + anyon classification. → me: continuing to Lyra Queue #5 = B3 geometric Langlands connection map.

— Lyra, B4 MTC construction from R-matrix v0.1. Substrate has all MTC ingredients: 66 simples (Phase B); Racah-Speiser fusion (E6/E10); R-matrix braiding (engine v0.3 / Toy 3617); ribbon element from Drinfeld R. At $q=2$ over $\mathbb{Z}$ (NOT root of unity case), modular S-matrix non-degeneracy is OPEN. Outcome A = substrate MTC = 3D TQFT (WRT) + anyon types = K-types + topological-matter substrate phases; Outcome B = pre-MTC = braided fusion + ribbon w/o full modular. Multi-week 66×66 S-matrix computation determines outcome.